

```

        android:textSize="25sp" android:padding="10dip"
        android:textColor="@android:color/black" />
        <LinearLayout android:id="@+id/x_axis"
            android:layout_width="match_parent" android:layout_
            height="wrap_content"
            android:background="@android:color/black">
            <TextView android:text="X-AXIS (+)ve:" android:layout_
            width="wrap_content"
                android:id="@+id/txtSensorValueLeft"
            android:textSize="30dip"
            android:layout_weight="1" android:layout_
            height="wrap_content"
            android:gravity="left"><TextView>
            <TextView android:layout_width="wrap_content"
            android:text="X-AXIS (-)ve:"
                android:textSize="30dip" android:layout_weight="1"
            android:id="@+id/txtSensorValueRight" android:layout_
            height="wrap_content"
            android:gravity="right" />
        </LinearLayout>

```

Here, you have laid out a single *TextView* to display the axis name, followed by two text views to display the positive and negative values of the axis, repeated for the Y and Z axes.

```

<TextView android:layout_width="match_parent" android:text="@
string/y_axis"
    android:layout_height="wrap_content"
    android:gravity="center"
        android:textSize="25sp" android:padding="10dip"
    android:textColor="@android:color/black" />
    <LinearLayout android:id="@+id/y_axis"
        android:layout_width="fill_parent" android:layout_
        height="wrap_content"
        android:orientation="vertical" android:gravity="center"
        android:background="@android:color/black">
        <TextView android:layout_width="fill_parent"
        android:text="Y-AXIS (+)ve:"
            android:layout_height="wrap_content"
        android:textSize="30dip"
            android:layout_weight="1" android:id="@+id/
        txtSensorValueUp"
            android:gravity="center" />
        <TextView android:layout_width="fill_parent"
        android:text="Y-AXIS (-)ve:"
            android:layout_height="wrap_content"
        android:textSize="30dip"
            android:layout_weight="1" android:id="@+id/
        txtSensorValueDown"
            android:gravity="center" />
    </LinearLayout>
    <LinearLayout android:layout_width="match_parent"
        android:layout_height="wrap_content"

```

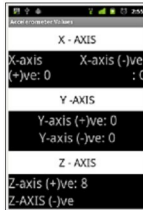


Figure 7: Values of the X, Y and Z axis when the phone was on the table

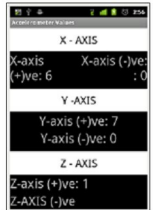


Figure 8: Values of the X, Y and Z axis when the phone was held in the hand

```

        android:background="@android:color/black"
        android:padding="5dip">
        </LinearLayout>
        <TextView android:layout_width="match_parent"
            android:text="@string/z_axis"
            android:layout_height="wrap_content"
            android:gravity="center"
            android:textSize="25sp" android:padding="10dip"
            android:textColor="@android:color/black" />
        <LinearLayout android:id="@+id/z_axis"
            android:layout_width="fill_parent" android:layout_
            height="wrap_content"
            android:orientation="vertical" android:background="@
            android:color/black">
            <TextView android:layout_width="wrap_content"
            android:text="Z-AXIS (+)ve"
                android:layout_height="wrap_content"
            android:textSize="30dip"
                android:layout_weight="1" android:id="@+id/
            txtSensorZValueUp" />
            <TextView android:layout_width="wrap_content"
            android:text="Z-AXIS (-)ve"
                android:layout_height="wrap_content"
            android:textSize="30dip"
                android:layout_weight="1" android:id="@+id/
            txtSensorZValueDown" />
        </LinearLayout>

```

After laying out the UI, you have to actually get access to the sensor and display the values it returns. First, reference the *UI* widgets in your Java class, so declare variables for each:

```

private SensorManager mSensorManager;
private Sensor mAccelerometer;
private TextView mSensorValuesLeft;
private TextView mSensorValuesRight;

```